

Ex: 1	Private Cloud using OpenStack
Date:	

Aim

This experiment focuses on setting up and configuring a private cloud infrastructure by deploying a single-node OpenStack instance using MicroStack. The objective is to gain practical knowledge of fundamental cloud services, including compute, networking, and storage resources, by launching and interacting with virtual machine instances within this environment.

What is OpenStack?

- OpenStack functions as an open-source, free standard for cloud computing.
- It is predominantly utilized to provide Infrastructure-as-a-Service (IaaS) solutions.
- This platform facilitates the on-demand provisioning and management of virtual machines (VMs).

What is MicroStack?

MicroStack provides a streamlined, single-node local execution of OpenStack. It serves as an excellent tool for educational purposes, experimental testing, or small-scale development. The platform simplifies the deployment and validation of OpenStack services on hardware ranging from laptops to compact servers.

Procedure

Step 1: Establish an Ubuntu VM within Oracle VirtualBox using these specifications: 8 GB RAM, 4 CPU Cores, and 30 GB of disk space.

Step 2: Complete the Ubuntu OS installation, then sign in with your configured credentials.
![Ubuntu Installation]

Step 3: Refresh the local package repository and upgrade existing software using the commands: 'sudo apt update' followed by 'sudo apt upgrade'.

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Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

thana@openstack-server:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
35 packages can be upgraded. Run 'apt list --upgradable' to see them.
thana@openstack-server:~$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14          linux-image-unsigned-7.0.0-14-generic  linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Upgrading:
base-files                libsgutils2-1.48          linux-firmware-netronome    packagekit                ubuntu-release-upgrader-core
fwupd                    linux-firmware-amd-misc    linux-firmware-nvidia-graphics  python3-distupgrade      ubuntu-server
gir1.2-packagekit-glib-1.0  linux-firmware-broadcom-wireless  linux-firmware-qlogic        sg3-utils                ubuntu-server-minimal
iproute2                  linux-firmware-intel-graphics  linux-firmware-qualcomm-graphics  sg3-utils-udev          ubuntu-standard
libfwupd3                  linux-firmware-intel-wireless  linux-firmware-qualcomm-misc    snapd                    update-notifier-common
libgcrngt20                linux-firmware-marvell-wireless  linux-firmware-realtek        ubuntu-kernel-accessories
libpackagekit-glib2-18     linux-firmware-mellanox-spectrum  mtd-news-config              ubuntu-minimal

Not upgrading yet due to phasing:
python3-software-properties  software-properties-common

Summary:
Upgrading: 33, Installing: 0, Removing: 0, Not Upgrading: 2
Download size: 571 MB
Space needed: 777 kB / 16.2 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 mtd-news-config all 14ubuntu6.1 [5,632 B]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 base-files amd64 14ubuntu6.1 [75.0 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 update-notifier-common all 3.207.1 [212 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libgcrngt20 amd64 1.12.0-2ubuntu1 [671 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 iproute2 amd64 6.19.0-1ubuntu1.1 [1,198 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-minimal amd64 1.570.1 [0.546 B]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-release-upgrader-core all 1:26.04.22 [24.9 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-distupgrade all 1:26.04.22 [102 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-standard amd64 1.570.1 [0.540 B]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 fwupd amd64 2.1.1-1ubuntu3.1 [12.7 MB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libfwupd3 amd64 2.1.1-1ubuntu3.1 [145 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libpackagekit-glib2-18 amd64 1.3.4-3ubuntu1.1 [121 kB]
Get:13 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 gir1.2-packagekit-glib-1.0 amd64 1.3.4-3ubuntu1.1 [25.5 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libsgutils2-1.48 amd64 1.48-3ubuntu3.1 [120 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-amd-misc all 20260319.git217ca6e4-0ubuntu1.1 [584 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-broadcom-wireless all 20260319.git217ca6e4-0ubuntu1.1 [11.2 MB]
122 [16 linux-firmware-broadcom-wireless 6,864 kB/11.2 MB 61%] 325 kB/s 28m19s

```

Step 4: Deploy the MicroStack environment via the beta channel using the command: 'sudo snap install microstack --beta'.

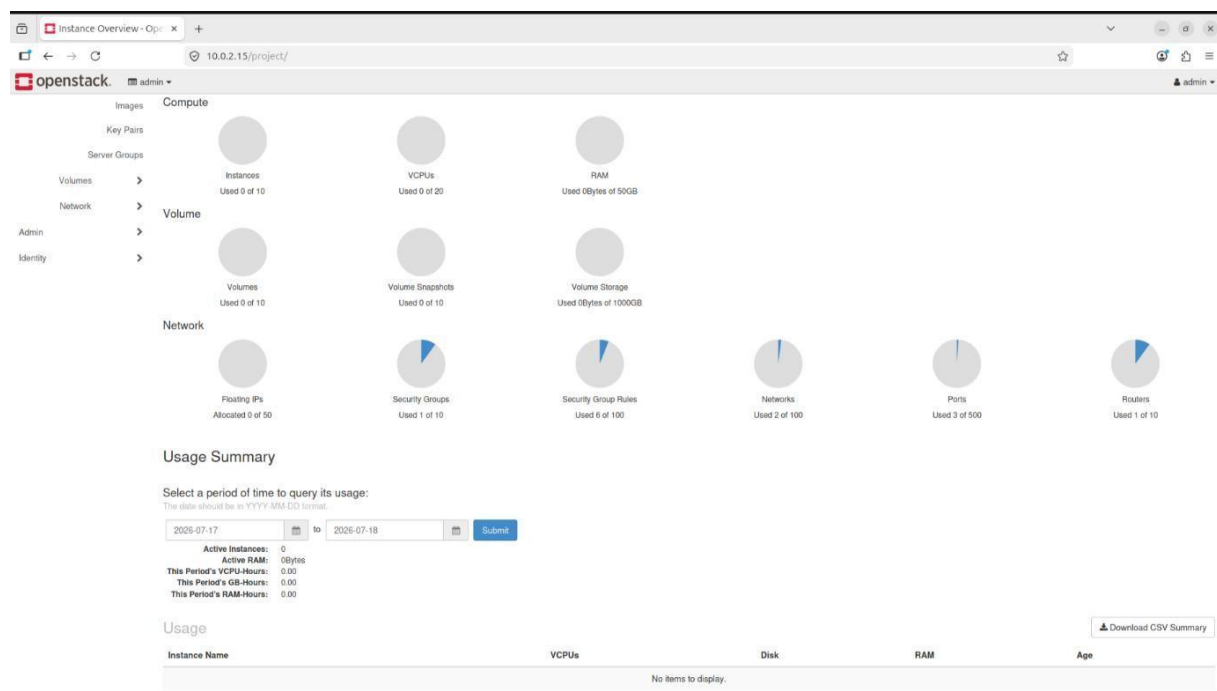
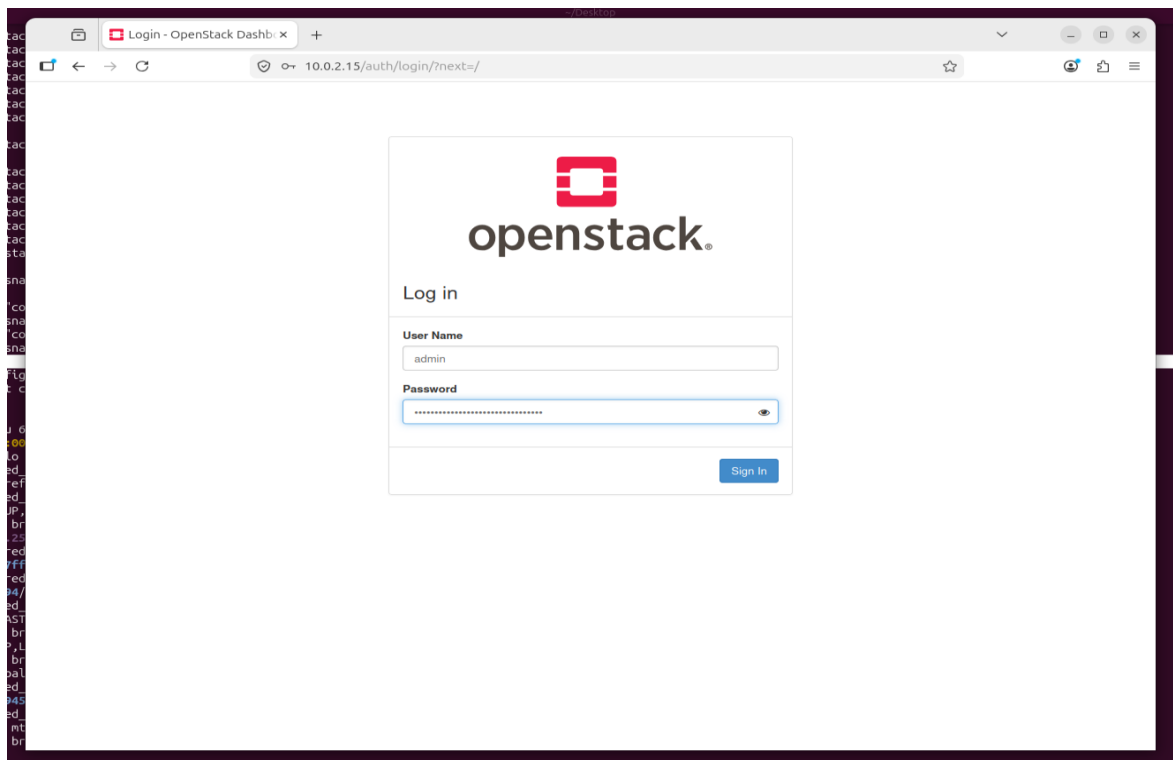
Step 5: Confirm the MicroStack installation status with: 'snap list microstack'.

Step 6: Execute the initialization process for OpenStack services by running: 'sudo microstack init --auto - -control'.

Step 7: Check the currently installed version of OpenStack: 'microstack.openstack --version'.

Step 8: Fetch the administrative Keystone password with this command: 'sudo snap get microstack config.credentials.keystone-password'.

Step 9: Access the OpenStack Horizon Dashboard via your web browser by navigating to the designated IP address displayed post-initialization. Use 'admin' as the username and the password retrieved in the previous step to authenticate.



Conclusion

The installation and configuration of MicroStack on Ubuntu were completed successfully, confirming that the Horizon Dashboard is fully operational.